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REVIEW

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Three-Dimensional (3D) cell culture monitoring: Opportunities and challenges for impedance spectroscopy

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Abstract

Three-dimensional (3D) cell culture has developed rapidly over the past 5–10 years with the goal of better replicating human physiology and tissue complexity in the laboratory. Quantifying cellular responses is fundamental in understanding how cells and tissues respond during their growth cycle and in response to external stimuli. There is a need to develop and validate tools that can give insight into cell number, viability, and distribution in real-time, nondestructively and without the use of stains or other labelling processes. Impedance spectroscopy can address all of these challenges and is currently used both commercially and in academic laboratories to measure cellular processes in 2D cell culture systems. However, its use in 3D cultures is not straight forward due to the complexity of the electrical circuit model of 3D tissues. In addition, there are challenges in the design and integration of electrodes within 3D cell culture systems. Researchers have used a range of strategies to implement impedance spectroscopy in 3D systems. This review examines electrode design, integration, and outcomes of a range of impedance spectroscopy studies and multiparametric systems relevant to 3D cell cultures. While these systems provide whole culture data, impedance tomography approaches have shown how this technique can be used to achieve spatial resolution. This review demonstrates how impedance spectroscopy and tomography can be used to provide real-time sensing in 3D cell cultures, but challenges remain in integrating electrodes without affecting cell culture functionality. If these challenges can be addressed and more realistic electrical models for 3D tissues developed, the implementation of impedance-based systems will be able to provide real-time, quantitative tracking of 3D cell culture systems.

KEYWORDS

3D cell culture, impedance spectroscopy, impedance tomography, in situ monitoring, multi-parametric sensing, real-time monitoring

1 | INTRODUCTION

As researchers have attempted to better replicate in vivo cell biology using in vitro systems, three-dimensional (3D) cell culture systems have developed rapidly (Baker & Chen, 2012; Edmondson, Broglie, Adcock, & Yang, 2014). One of the key goals of 3D cell cultures is to reproduce the spatial organization and function of cells in the human body. There are a variety of formats including spheroids, organoids, and engineered tissues. In 2D, a cell becomes polarized as surface integrins enable it to attach to the substrate. In 3D, the cell surface integrins are distributed more randomly as the cell attaches to the scaffold/matrix in multiple directions. The physiological differences induced by the different attachment modes and the resulting cell/media interactions mean that cellular functions such as proliferation, differentiation, survival, and mechanical signaling vary between 2D and 3D cell cultures. The details of these variations have been reviewed extensively (Duval et al., 2017),

and many studies have shown variations in metabolism between 2D and 3D systems (Brajša, Trzun, Zlatar, & Jelić, 2016; Verjans, Doijen, Luyten, Landuyt, & Schoofs, 2017) leading to a focus of these systems for early-stage drug testing and disease progression.

As experiments migrate from 2D to 3D cultures, traditional monitoring techniques need to evolve to accommodate the needs of researchers and the specific questions being asked of the 3D systems. Currently, it may take up to 30 days to grow a fully differentiated and mature 3D tissue such as skin (Carlson, Alt-Holland, Egles, & Garlick, 2008; Lee et al., 2014). During this time and after it has matured, researchers need to ensure the tissue has grown correctly and is biologically sound for experimental use. However, at this point in time, there are few convenient, bench-top platforms to monitor this process without physically disrupting the model. In 2D culture, bright field microscopy is often used in real-time to monitor cell growth, but this is generally not possible with nontransparent 3D tissues. To date, the most common techniques used to visualize and validate 3D models are end-point histology and fluorescence microscopy (Dmitriev, 2017). Even though these techniques are gold standards, they are time consuming, and in most cases, require the destruction of the sample for sectioning and staining. While live cell approaches abound, reporter constructs and stains have a limited lifespan and are not targeted to 3D cultures (Walker-Daniels, 2012). Introducing new standardized methodologies to monitor 3D cultures in real-time would save time, reducing research cost and providing a quality assurance pathway for manufacturing.

Cell visualization is the most intuitive way of identifying cell health, but a range of other methods have been developed to indirectly assess cells in culture. Physical and chemical probes that measure the levels of oxygen, glucose, CO_2, and pH of cell culture media can determine cell metabolism and health status (Bavli et al., 2016; Hossein Mahfouzi, Amoabediny, Doryab, Hamid Safiabadi-Tali, & Ghanei, 2018; Jenkins, Dmitriev, Morten, McDermott, & Papkovsky, 2015; Shaibani, Etayash, Naicker, Kaur, & Thundat, 2016; Weltin et al., 2014; Weyand et al., 2015). Traditional techniques that measure parameters such as glucose and oxygen concentrations take advantage of ion-permeable membranes and selective enzymes to determine concentration levels (Oliver, Toumazou, Cass, & Johnston, 2009). However, a broad range of transducers can also be used to indirectly measure these parameters of cell health (Modena, Chawla, Misun, & Hierlemann, 2018). Field effect transistors (FETs) and light-addressable potentiometric sensors (LAPs) have emerged as useful sensors capable of detecting cellular metabolic products and provide effective spatial resolution within 2D cell culture systems. FETs are generally comprised of an ion-permeable membrane that is integrated between the source and drain to allow for measurement of extracellular ion concentrations and cellular respiration, while LAPs allow for pH measurements of cells by recording changes in photocurrent (Curto et al., 2017; Dantism, Takenaga, Wagner, Wagner, & Schöning, 2015; Poghossian, Ingebrandt, Offenhäusser, & Schöning, 2009). Photocurrent changes measured by LAPs have also been used to monitor wound healing 2D cultures (Wagner et al., 2016). Most of these culture characteristics are measured in the media downstream of cells and they give real-time information without destruction of the culture or tissue. These sensors are commercially available and are usually incorporated within bioreactor systems that are used to grow and monitor spheroids or cells in suspension over time (Alexander, Eggert, & Wiest, 2018; Weyand et al., 2015). This approach assumes that the media is reflecting the behavior inside the tissue. With 3D cultures such as spheroids, this approach may become problematic as spheroids display diffusion limits to nutrients leading to cell death at the center of the culture (Mehta, Hsiao, Ingram, Luker, & Takayama, 2012). Further, coculture systems can lead to local variations in behavior (Goers, Freemont, & Polizzi, 2014), which are difficult to detect in the whole system by media sampling.

An alternative approach involves the characterization of electrical properties of cells in culture. Trans-epithelial electrical resistance (TEER) measurements are widely used in biology and tissue engineering to determine epithelial and endothelial cell health and membrane integrity. These measurements give an indirect, noninvasive assessment of the permeability of cellular tight junctions and ultimately the barrier function of a cellular monolayer or 3D tissues (Chen, Einspanier, & Schoen, 2015; Schmitz et al., 2018). There are several variables known to affect the final readout including electrode size, measurement temperature, media formulation, time in culture, and cell passage number, making it difficult to standardize measurements (Elbrecht, Long, & Hickman, 2018; Srinivasan et al., 2015). A detailed review of TEERs and a guide to laboratory standardization is presented by Srinivasan et al. (2015).

Derived from the same principles as TEER, electrical impedance spectroscopy (EIS), is a well-established tool for monitoring 2D cell cultures. EIS measurements are acquired with electrodes positioned beneath the culture and changes in current/voltage through the cells are measured as a frequency sweep is undertaken. This process allows the electrical resistance and capacitance of the cells to be monitored (Benson, Cramer, & Galla, 2013; Groeber et al., 2015). While TEER measures currents at one frequency, sweeping the frequency allows EIS to obtain more detailed information about cellular health status over time. Currently, there are a range of commercially available TEER and EIS devices used for assessing cell barrier function, integrity, and cell proliferation in real time. These include OrganoTEER (MIMETAS, Leiden, The Netherlands), xCELLigence Real Time Cell Analysis (ACEA Biosciences, Inc., CA), EVOM2 (World Precision Instruments, FL) and Millicell-ERS2 (EMD Millipore Corporation, Billerica, MA). However, these devices are mainly designed for monitoring adherent cells grown in 2D or transwell-based coculture systems. Electrodes have also been integrated into a range of different 3D cell culture models and microfluidic platforms for EIS monitoring (Curto, Ferro, Mariani, Scavetta, & Owens, 2018; Lee et al., 2016; Lei, Liu, & Tsang, 2018). In particular, cell proliferation within spheroids have been successfully monitored using this technique (Bü, Diener, Frey, Kim, & Hierlemann, 2016; Lei, Lin, & Tsang, 2017; Lei, Liu, & Tsang, 2018; Pan et al., 2019).

In addition, EIS has also been incorporated into 3D culture systems for multiparametric monitoring including bioelectrical sensing and the detection of cell metabolites. Despite the need for the integration of in-line sensing and optical monitoring tools for

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real-time and rapid biological readouts, little research exists where these have been successfully integrated. This is especially evident with highly modular, sensor-based, organ-on-chip systems that are difficult to integrate with current microscopy approaches due to design incompatibilities (Curto et al., 2017; Esch, Ueno, Applegate, & Shuler, 2016; Zhang et al., 2017). Multiparametric approaches incorporating sensing systems, optical readouts, and techniques that can provide better spatial resolution are currently needed. This review focuses on the varied methods that have been used, the challenges and limitations associated with different strategies, and the factors that need to still be addressed to enable real-time, quantifiable results from these systems.

2 | THEORETICAL BACKGROUND OF IMPEDANCE SPECTROSCOPY

To measure the EIS of cell cultures, often a frequency-dependent small sinusoidal voltage (V(ω)) is applied through the cells (or tissue) and the changes in the resulting sinusoidal current (I(ω)) between the electrodes are recorded (Benson et al., 2013; Elbrecht et al., 2018; Srinivasan et al., 2015). Using Ohm's law, it is then possible to determine the electrical impedance (Z(ω)) of cell cultures using the following equation:

$$Z(\omega) = \frac{V(\omega)}{I(\omega)} = Z_{RE} + iZ_{IM}(\omega). \tag{1}$$

Equation (1) shows the resulting frequency-dependent impedance where Z_{RE} is the real part of the impedance components and Z_{IM} (ω) is the imaginary part. Z_{RE} is usually related to the resistive behavior of the system, which is independent of the frequency, while Z_{IM} (ω) is related to a more capacitive behavior of the system, which is frequency dependent. These components can be related to the magnitude (|Z (ω)|) and phase (θ(ω)) of the output signal and their relationships are shown in Equations (2) and (3).

$$|Z(\omega)| = \sqrt{Z_{RE}{}^2 + Z_{IM}(\omega)^2}, \tag{2}$$

$$\theta(\omega) = \tan^{-1}\left(\frac{Z_{RE}(\omega)}{Z_{IM}(\omega)}\right). \tag{3}$$

Depending on the research question, a combination of, or parts of the aforementioned components are used to assess the presence and/or changes to cells in cell culture systems. Others use normalized impedance data to express cell system changes such as cell proliferation or death using a cell index (CI). The CI allows analysis of the data without taking into account the impact of electrode–electrolyte combination and parasitic elements and it is defined in Equation (4) (Pan et al., 2019).

$$\mathrm{CI} = \frac{|Z(\omega)| - |Z(0)|}{|Z(0)|}. \tag{4}$$

Several equivalent circuits have been designed using the components of impedance to model and quantify the electrical properties of cells grown in 2D systems. Figure 1 shows the equivalent circuit generally used to describe live cells in a monolayer (Benson et al., 2013). This circuit takes into account the resistance of the solution or electrolyte (phosphate buffered saline [PBS] or cell media) R_{medium}, the resistance and capacitance of the cell monolayer that contributes to the total spectrum R_{TEER} and C_{CI}, respectively, and the capacitance of the electrodes and electrode-medium interphase C_{EI} (Benson et al., 2013). As the cells spread, proliferate, migrate, or die across the electrodes, changes can be observed in the impedance signal and in the values of the different components of the circuit. The correlation of the impedance signal with the physical changes of the cells, electrical equivalent circuit, and electrode setup for 2D adherent cells are well developed in EIS and comprehensive reviews of impedance measurements of cells are available and demonstrated (Benson et al., 2013; Canali et al., 2016; Elbrecht et al., 2018; Srinivasan et al., 2015).

3 | IMPEDANCE MEASUREMENTS IN 3D CELL CULTURE

The implementation of EIS in 2D cultures relies on cells attaching to the electrodes. This does not occur in 3D cultures as they contain matrices such as biopolymers (collagen, fibrin), polymer scaffolds, or discrete suspended cell clusters. As a result, there are a number of areas that the EIS system needs to be adapted for 3D applications:

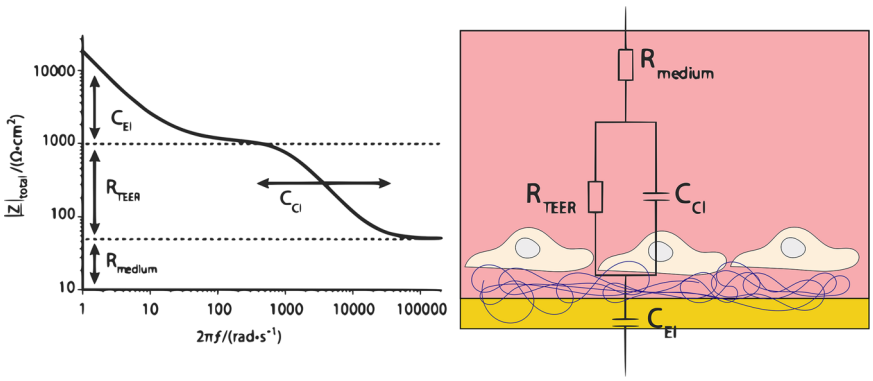

FIGURE 1 Equivalent circuit and typical impedance spectrum of cells in 2D. Adapted with permission (Benson et al., 2013)

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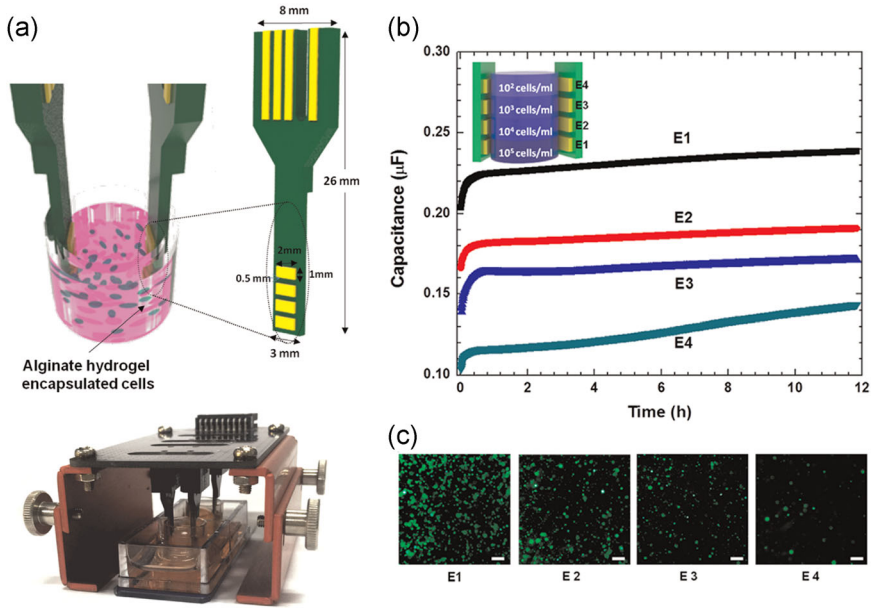

FIGURE 2 (a) Vertical electrodes to measure the capacitance of cells in a hydrogel with vertical resolution. (b) Real-time capacitance of hydrogels seeded with different cell concentrations. (c) Fluorescent optical images of hydrogels containing different cell densities. (scale bar: 100 μm). Reproduced with permission (Lee et al., 2016). Copyright 2015, Elsevier B.V.

1. Ensuring sensitivity of the electrode array to cells suspended in or attached to the matrix/scaffold by ensuring there is an even electric field through the whole system.
2. Adaptable to a range of different 3D cell culture formats—systems are often designed to address a specific biological research question or tissue function and so their architecture varies.
3. Ensuring that electrodes do not influence cell responses or change the mechanical environment in 3D cell culture systems.

The design and integration of electrodes within culture systems appears to be central to the effective monitoring of cells within scaffolds and hydrogels. It has been demonstrated that the electrical properties of cells can be monitored when they are immersed within a hydrogel rather than directly attached to a planar electrode (Lee et al., 2016). Lee et al. (2016) incorporated a matrix of vertical electrodes (Figure 2a) that were able to monitor the migration, proliferation, and apoptosis of cells through a gel matrix in 3D. In this system, cells were grown in an alginate hydrogel scaffold and vertical parallel electrodes were used to measure the capacitance of the system. An increase in capacitance was shown to be proportional to an increase in seeded cell density within the hydrogel (Figure 2b,c), showing that by measuring capacitance of the system, it is possible to monitor cell number. To monitor cell proliferation and migration, one frequency (1 kHz) was used to obtain the capacitance of the system rather than the whole impedance spectrum. The chosen frequency coincides with the frequency typically used to monitor cells in 2D (Benson et al., 2013; Srinivasan et al., 2015), but the study did not explore how cell death or migration affects the whole spectrum or whether other changes in the spectrum could be correlated to these cellular processes.

In many instances, cells are grown in a 3D spheroidal structure in which the cells clump together and proliferate. These models are used extensively in cancer research to understand drug treatments for tumors. Lei et al. (2018) used in-plane electrodes to monitor the viability of spheroids developed in an agarose hydrogel using the hanging drop technique (Figure 3a). Changes in the impedance magnitude was successfully correlated with different cell numbers seeded in the hydrogel (Figure 3b). It is interesting to note here that while the authors chose the standard 1 kHz as the optimal sensing

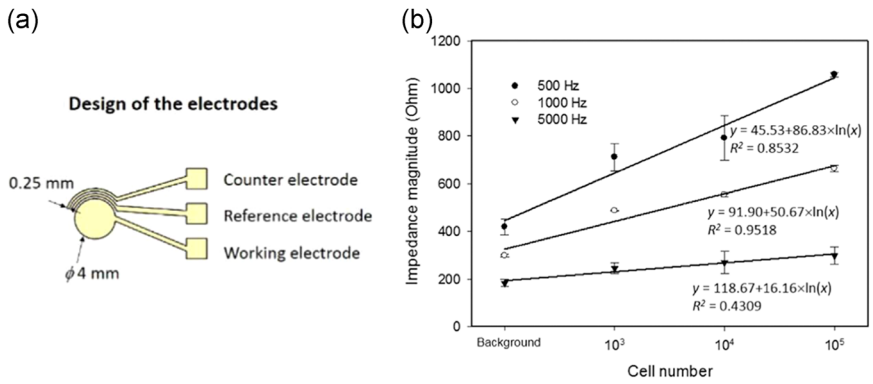

FIGURE 3 (a) Horizontal electrodes for measuring cells in hydrogel using the hanging-drop technique. (b) Correlation between cell number and impedance magnitude measured at discrete frequencies (500, 1000 and 5000 Hz). Reproduced with permission (Lei et al., 2018). Copyright 2017, Elsevier B.V.

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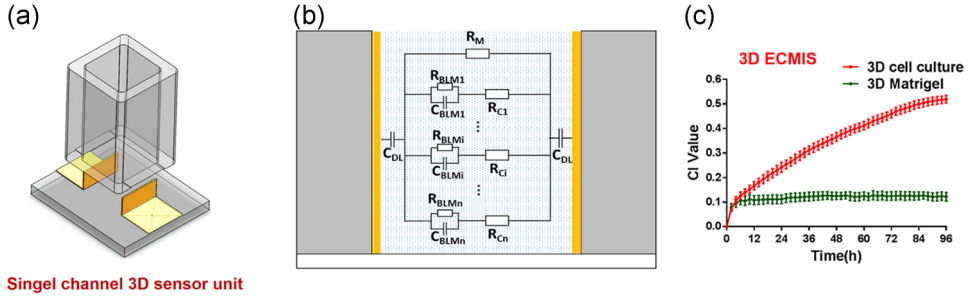

FIGURE 4 (a) Schematic of the parallel electrodes for the impedance biosensor. (b) Equivalent circuit of cells in 3D within the matrigel. (c) Measurement of cells in 3D culture using the parallel electrodes (3D ECMIS). Reproduced with permission (Pan et al., 2019). Copyright 2018, Elsevier B.V.

frequency, as was seen in the earlier study, there were larger variations in the impedance magnitude at lower frequencies. These variations were accounted for by the authors as being related to the increases in the noise as the frequency is reduced, due to noise being proportional to 1/f.

The position of the electrodes within 3D culture systems (parallel vs. in-plane) has also been shown to affect detection outcomes (Xu et al., 2015). Pan et al. (2019) found that in-plane electrodes were not able to significantly detect the proliferation of spheroids in a matrigel, whereas parallel electrodes successfully monitored their proliferation over time (Figures 4a and 4c). This data correspond with the theory that parallel electrodes can achieve more sensitive detection due to a more uniform electric field (Xu et al., 2015). Nevertheless, this challenges the results from the earlier studies from Lei et al. (2018) using in-plane electrodes. Factors influencing these differences may be gel thickness and electrode design. When in-plane electrodes are used in a 3D system, the applied current can only penetrate a small volume and distance, whereas parallel electrodes force the applied current to go across the entire sample. In this instance, it is to be expected that more signal can be detected and signal to noise increases as the current penetrates a larger volume of the 3D culture.

While the integration of standard metal electrodes into 3D culture is challenging due to the stiffness of the electrode materials and the need to connect them back to the outside world, one group has been exploring the application of conductive polymers as both the cell scaffold and the electrodes (Khan et al., 2019). As the cells adhere to this conductive scaffold, its impedance changes enabling cell adhesion to be detected (Figures 5a and 5c; Del Agua et al., 2018). Interestingly, the optimal sensing frequency found using a poly(3,4-ethyenedioythiophene): xanthan scaffold was between 0.1 and 100 Hz (Figure 5b). As mentioned in the previous example, noise varies with 1/f, so at lower frequencies we would expect that noise would increase significantly. The potentiostat used for the impedance measurements may be able to compensate, however, little information is given by the authors of this study to determine if this approach has been implemented. Acquisition rate may also be problematic at low frequencies, though in the case of cell monitoring, this may not be an issue since these are long-term cultures of days and weeks rather than minutes and hours.

In 2D, it is well established that low frequency signals are dominated by the double layer capacitance of the electrodes (Benson et al., 2013). Nevertheless, in this study, the electrode properties change as the cells are growing within the conductive hydrogel and it is consistent with other studies using conductive scaffolds (Inal et al., 2017). Further, modifying the properties of the 3D scaffolds (physical and chemical properties) should be done with caution as scaffold properties can have a big influence on cell behavior and function

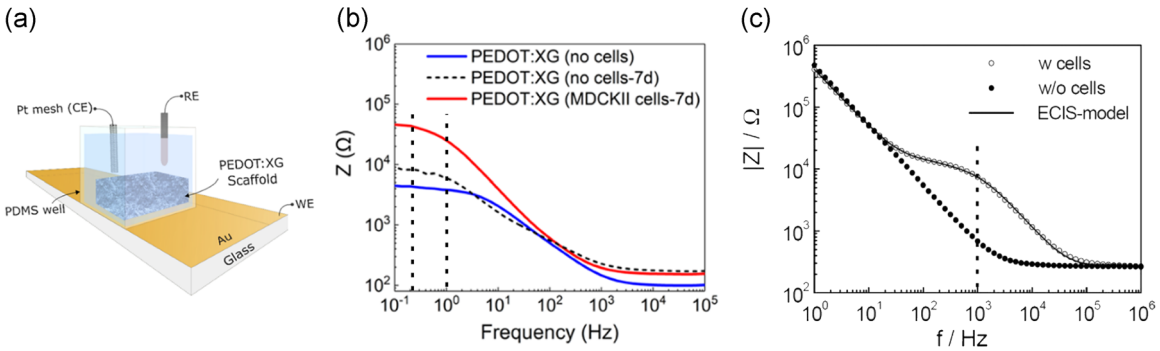

FIGURE 5 (a) Schematic of the poly(3,4-ethylenedioxythiophene) xanthan gum (PEDOT: xanthan) electroactive scaffold used to host and monitor cells. (b) Impedance spectrum of the scaffold with no cells (blue), after cell growth within the scaffold over 7 days (red) and PEDOT: xanthan gum scaffold after 7 days of incubation (37°C, 5% CO_2) in culture media without cells (dashed lines). Reproduced with permission (Del Agua et al., 2018). Copyright 2018, American Chemical Society. (c) Typical impedance spectrum of 2D cells grown on electrodes. Reproduced with permission (Xu et at 2015) Copyright 2015, Elsevier B.V.

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(Bahcecioglu, Hasirci, & Hasirci, 2019; Caliari & Burdick, 2016; Sridharan, Ryan, Kearney, Kelly, & O'Brien, 2019) and these influences need to be understood and cross-validated with other characterization tools before implementation. This suggests that the nature of the scaffold, its potential use as a working electrode and the nature of the 3D model (spheroids or tissue) plays a significant role in the final outcome of the measurements and as such models are needed to compare results with theoretical systems.

In undertaking this review, we could find only one study that has directly compared equivalent circuits for 2D and 3D systems (Pan et al., 2019). They demonstrated that a matrigel matrix resistance component was in parallel with the electrical components representing cell impedance (Figure 4b). This in contrast to a typical 2D equivalent circuit, which shows that the media component is in series with the components representing cell impedance (Figure 1). This is explained by looking at the fundamental differences in cell interactions occurring in the two different models. In 2D, the cells reside below the media (series circuit), while in 3D, the cells reside within the scaffold, which resembles more of a parallel circuit. Furthermore, Pan and colleagues paired simulations derived from their 3D equivalent circuit with their experimental data using epithelial cells and found that the presence of cells in 3D decreased the overall impedance of the system. This was due to an increase in gel conductivity after the addition of cells, which was attributed to the presence of gap junctions between cells that allowed electrical connection between neighboring cell membranes (Pan et al., 2019). However, other studies with similar cell types have shown the opposite effect, with the introduction of cells increasing the overall impedance of 3D systems (Inal et al., 2017; Lee et al., 2016; Lei et al., 2018).

Groeber et al. (2015) are among the only published groups using the whole EIS spectrum and information provided by an equivalent circuit to monitor tissue growth in a coculture system. They used parallel electrodes to measure the impedance of an artificially grown epidermis at different levels of maturity as shown in Figure 6b. They developed an equivalent circuit in which there are a series of connections of *n* parallel resistor-capacitor circuits, which correlated with a different maturity phase of the artificially grown epidermis (see Figure 6a). The data obtained was validated against an isolated human epidermis showing that the artificially grown epidermis at days 9–12 had the same electrical characteristics as the human epidermis. In the study, 1 kHz was found to be an optimal sensing frequency, suggesting that it may well be the nature of the conductive scaffold that reduced the frequency in the study by Agua et al. (2018). To measure the impedance of the tissue, Groeber et al. (2015) grew the tissue in a transwell and transferred the transwell to an impedance platform every time a measurement was required. It is possible to use ideas from the literature to envisage how electrodes could be inserted into the transwell to monitor the system in situ or within a microfluidic device as undertaken by Sriram et al. (2018). A natural extension of this approach would be to design electrodes, so they are small enough to contact only one layer of tissue construct. This could enable the extraction of information of cells within a select region and monitoring of the individual features as the tissue grows. While Groeber and colleagues monitored growth and differentiation of epidermal cells, electrodes placed at different layers of a full thickness skin equivalent may allow for multicellular monitoring as well as the potential monitoring of skin processes such as wound healing and barrier function. Currently, skin tissue maturity is assessed by histology and immunostaining at different time points,

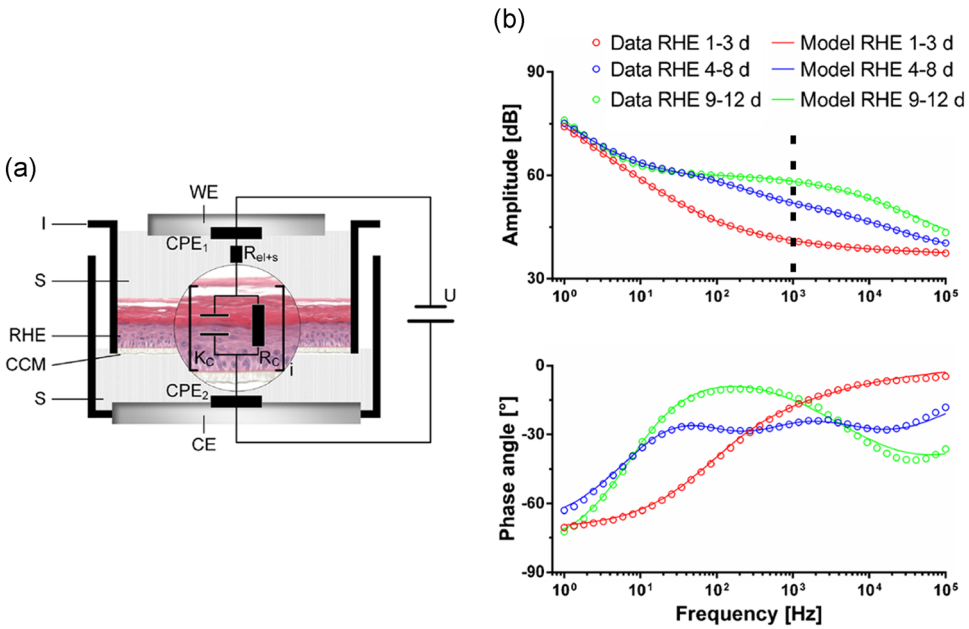

FIGURE 6 Parallel impedance electrodes to measure the maturity of a reconstructed human epidermis (RHE). (a) Detailed view of one measurement chamber with a RHE in a transwell insert and equivalent circuit. (b) Bode plots of impedance spectra from RHE at different time points. Reproduced with permission (Groeber et al., 2015). Copyright Groeber et al. (2015)

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so a layer-by-layer electrode setup which monitors tissue impedance as it grows could allow researchers to track growth and cellular processes quickly and without destroying the model.

4 | MULTIPARAMETRIC MONITORING OF 3D CULTURES

Impedance changes are a compilation of multiple processes occurring in cell culture systems. In case of 2D cultures, significant impedance changes have been successfully correlated with cell proliferation and death. However, in 3D cultures, cellular processes are more complex, thus impedance changes may be influenced by different factors such as matrix degradation, cell spreading, migration, or proliferation. This raises the need for a secondary or multiparametric approach to further understand, monitor, and validate 3D culture systems. In-line monitoring systems coupled with impedance can lead to better real-time and multiparametric biological understanding of complex 3D systems. Zhang et al. (2017) developed a multisensor-integrated organ-on-chip platform that contains two organ modules, one physical/chemical sensing module and one bioelectrochemical sensing module, all connected via a perfused system. The physical/chemical sensing system consisted of optical pH and oxygen sensors. In addition, gold electrodes functionalized with desired antibodies for specific detection of molecules in the media could be incorporated into the bioelectrochemical sensing module. The EIS changed upon binding of the target molecule and detection of different concentrations of the analyte/biomarker was possible. They found that the amount of biomarker that could be captured by the functionalized electrode surface was proportional to the concentration in the solution, with detection levels of 0.1–100 ng/ml, similar to many commercial enzyme-linked immunosorbent assay systems. Thus, biologically relevant concentrations were detected. Even though this is a complex system, it contains fundamental parameters for real-time monitoring of organoids. However, this system is highly modular and specific to the microfluidics system they utilize. This makes it harder to standardize and integrate these systems with current tissue culture consumables and microscopes. Nevertheless, to date, this is the most complete multiparametric monitoring system for organoids.

Others have also used a multiparametric monitoring approach coupled with electrical measurements to monitor 3D cell cultures. Misun, Hierlemann, and Frey (2018) correlated amperometric (current) changes to glucose and lactate concentrations in spheroids, giving real-time information on the metabolic state of these microtissues. For the detection of glucose and lactose in these spheroids, a microfluidic polydimethylsiloxane chip with an enzyme-based hydrogel coated onto sensing electrodes was proposed. However, the sensor chip and well was limited in its use as it is specifically designed for spheroid growth via the hanging-drop culture technique. The sensor also depends on the effective release of glucose from the spheroid to the media. Another approach to monitor glucose concentrations and cell proliferation was shown by Curto et al. (2017). They developed an in vitro cell monitoring platform, which monitored the impedance changes of tissue by means of an organic electrochemical transistor (OECT) at the bottom of the well. Glucose was also monitored with an OECT-based glucose biosensor, but the glucose biosensor was not in situ. Media was collected at the outlet of the perfused system and later measured with the sensor. This multiparameter system allowed the monitoring of kidney cell metabolic state via the glucose sensor as well as changes in resistance and capacitance in real-time, while the kidney cell epithelium recovered from an electrical wound.

All of the above systems allow for real-time monitoring of the metabolic activity of cells through measuring glucose, pH, and oxygen levels in the media. Even though they are sensitive, they depend on the efficient release of the molecule of interest into the media, which can be limited in a 3D cell culture system. The lack of spatial resolution displayed by most of the monitoring techniques within these devices remains a drawback. Correlating metabolite levels in media with a technique that gives information about spatial resolution would allow for more specialized biological information to be gained from the monitoring system.

5 | 3D IMPEDANCE TOMOGRAPHY

In 3D cell culture, obtaining spatial resolution is of interest as it allows researchers to determine the location of cells over time, monitor the specific cell health, and helps to highlight differences in cell health and function in different areas of the tissue. Electrical impedance tomography (EIT) is a relatively new technique that has emerged as a potential solution to provide better spatial resolution within 3D tissue cultures. However, most recent efforts in EIT have been focused on its use in vivo and as a medical diagnostic, where electrodes are placed on the body to obtain an image and health status of an organ of interest (Fernandez-Corazza et al., 2018; Schwartz, Chauhan, & Sadleir, 2018; Sun, Yue, Hao, Cui, & Wang, 2019).

Similar to the implementation of EIS, implementing EIT involves the application of a small current to a sample, such as the human body, tissue model, or inanimate objects with different conductive properties (generally used for prototyping), and the voltage is measured to obtain the electrical properties of the sample (Chitturi & Nagi, 2017). Different electrode designs, current injection strategies, and image reconstruction algorithms have been recently reviewed by Chitturi and Nagi (2017). In general, the set up involves a matrix of electrodes that is placed around the sample, a current is injected through a pair of electrodes while the voltage changes are measured through a different pair of electrodes. The values obtained are then used to calculate the resistivity of the sample and reconstruct the image using the chosen algorithm.

The successful use of EIT as a medical diagnostic and its ability to obtain in vivo information, suggests that it may be an effective monitoring technique for future 3D cell cultures. Researchers have demonstrated the measurement of 3D tissue impedance and

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subsequently reconstruction of the respective images using EIT (Jamil et al., 2016; Wu, Yang, Bagnaninchi, & Jia, 2017; Wu, Yang, Bagnaninchi, & Jia, 2018; Wu, Zhou, Yang, Jia, & Bagnaninchi, 2018; Yang et al., 2017; Yin, Wu, Jia, & Yang, 2018; Yin, Yang, Jia, & Tan, 2017). For monitoring 3D cell aggregates or spheroids, two electrode distribution designs were implemented. In one study, circular electrodes were radially distributed at the bottom of a well, in which case current was injected through a pair of electrodes and voltage was measured at the adjacent pair, while the reference electrode was connected to ground (Figure 7a). In another study, current was applied to rectangular electrodes placed at the boundary of the well and voltage was measured as shown in Figure 7b. Using both electrode designs, the authors were able to monitor the presence and growth of spheroids (Yin et al., 2018) as well as spheroid disintegration upon the addition of Triton-X 100 (Figure 7c). However, the resolution of the output is not yet at the cellular level as shown by the heat maps in Figure 7c. Nevertheless, compared to EIS, additional information can be gathered using EIT including the position and potentially the size of spheroids and tissues in culture systems. Given the issue of contraction in some collagen-based 3D models, as well as the presence of contraction in wound healing models (Lotz et al., 2017), a potential application for EIT could be to monitor contraction over time or after wounding in these models. Others have designed similar set ups to those demonstrated in Figure 7, with the eventual aim of monitoring cell cultures, but have not yet tested it using cells (Xu et al., 2018).

Likewise, a more complex setup of electrodes was designed with the intention of monitoring 3D cultures using EIT (Lee et al., 2014). The quality of the designed device (Figure 8) was assessed using performance metrics such as crosstalk, amplitude stability error, total harmonic distortion, and signal to noise ratio. While this system was not tested using cells, the high quantity of electrodes positioned all over the well (Figure 8) could potentially increase the final image resolution and subsequently biological output, as more data points may be obtained from the well. Conversely, having electrodes in such close proximity may also increase the crosstalk and artefacts such as stray capacitance. A fine balance between the number of electrodes and electrode design is needed to obtain optimal resolution in the presence of 3D cell cultures.

The above studies have demonstrated a prospective use for EIT in 3D cell culture monitoring, however current applications do not offer a clear advantage over EIS. EIS is comparatively simpler and may be more cost effective in obtaining the electrical properties of a biological 3D sample. While EIT may be an attractive solution for better spatial resolution in the monitoring of 3D cultures, the current sensitivity obtained can only give qualitative information at the tissue level as opposed to information at the cellular level. Achieving resolution at the cellular level will give a better indication of what is

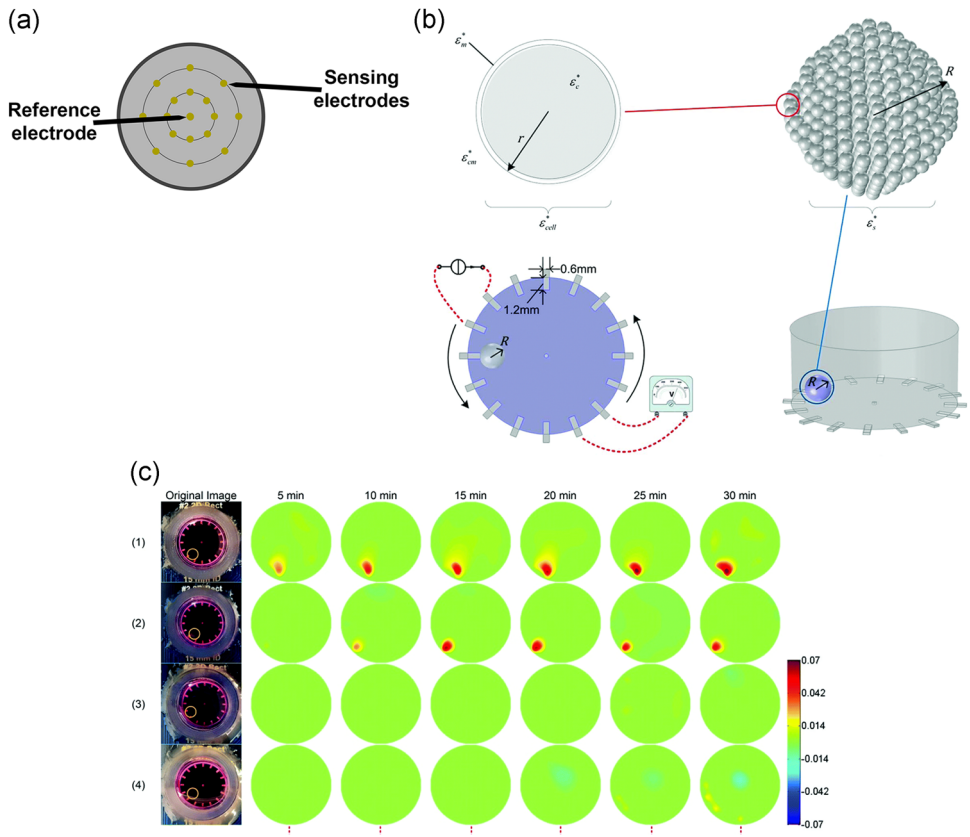

FIGURE 7 (a) Radially distributed circular electrodes (adapted from Yin et al., 2018). (b) Rectangular electrodes at the boundary of the well were used to monitor spheroid and (c) reconstructed images for the spheroid samples in 2% Triton X-100 solution (1-2) and HG culture medium (3–4). Reproduced with permission (Wu, Yang et al., 2018; Wu, Zhou et al., 2018). Copyright Wu et al. (2018)

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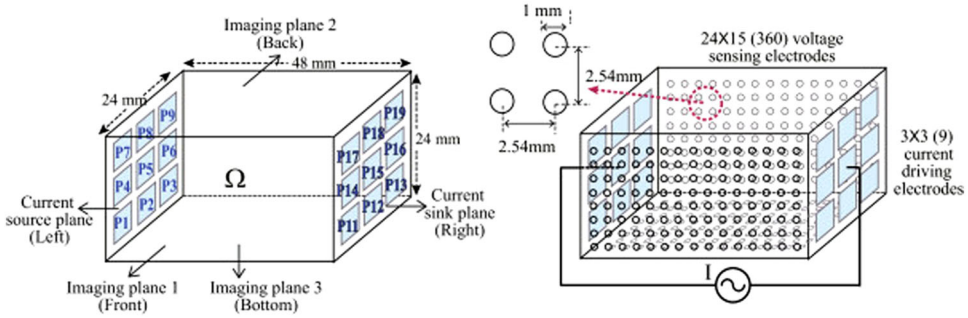

FIGURE 8 The electrode design shows the placement of current injecting electrodes on parallel walls and the placement of voltage sensing electrodes on the bottom surface as well as the two parallel walls. Reproduced with permission (Lee et al., 2014). Copyright Lee et al.; licensee BioMed Central Ltd. 2014

occurring as cell cultures grow over time, after treatments and within different regions of the 3D culture.

6 | FUTURE DIRECTIONS

Three-dimensional culture is expanding across research and industrial laboratories, as it offers the possibility of more physiologically relevant models for drug testing and replication of diseased tissues. Therefore, there is a need to characterize and standardize monitoring techniques across different laboratories. This will increase the scope of 3D culture as well as ease its implementation (Verjans et al., 2017). Further, it will allow researchers to select matured tissues that are physiologically relevant in their contexts and allow them to perform experiments in a more controlled environment where variability within cultures can be monitored. However, a lack of monitoring techniques represents a limitation for further growth. Developing monitoring techniques targeted for 3D cultures will allow for easier implementation and standardization of these cultures. The development of reproducible, scalable, and validated 3D cell culture systems that can handle a diverse range of cell types and cocultures while providing real-time feedback on culture health will improve our fundamental understanding of biology. The ability to monitor the cultures as they grow and assess their viability before testing them with an external stimulus will accelerate the rate of discovery.

Multiple approaches to integrating electrodes into 3D cell culture systems have enabled the monitoring of cell health over time. To date, researchers have developed in situ sensors using EIS to monitor cell proliferation and viability. Furthermore, they have created devices with functionalized electrode surfaces to enable the measurement of the metabolic activity of 3D cell cultures. However, each of these devices is highly customized for the specific application and 3D culture technique, making standardization difficult. In future, EIT may provide better spatial resolution, but the currently achieved resolution remains low and the data gathered can also be obtained with EIS, which utilizes simpler hardware and data analysis. In addition, more research is needed to characterize and validate the effects of matrix composition of different matrices, including hydrogels, on both cell structure and function as well as the effect on the corresponding impedance response. In this regard and to the best of our knowledge, only one group has correlated impedance changes upon changes to hydrogel composition (Inal et al., 2017). It is still unclear how the system is changed upon the addition of cells to different nonconductive hydrogels.

An ideal monitoring system for 3D cell culture would be able to monitor multicellular models in situ and in real-time. Having an ideal spatial resolution at the cellular level (~50 μm) will allow monitoring of specific cells in a multicellular construct and their state within the system (live, death differentiated). Impedance alone can achieve real-time monitoring of cell growth and proliferation, however, it should be complemented and correlated with measurements from the monitoring of other cell functions such as metabolic activity, which can be detected downstream in the media using functionalized surfaces as transducers or sensors of interest.

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